

ASSIGNMENT-5

Task – 1: Understanding Users and Groups

SOLUTION-

Linux User Model

1. Root User

- Root is the superuser in Linux.
- It has full control over the system and can perform all administrative tasks.
- Root user has UID 0.

2. System Users

- System users are created by the operating system and services.
- They run background processes such as web servers and databases.
- Examples: daemon, nobody, www-data.

3. Regular Users

- Regular users are created for people who use the system.
- They have limited permissions for security purposes.
- Example: praveen.

Structure of /etc/passwd

Command-

```
cat /etc/passwd
```

```
praveen@praveenasus:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101:/nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103:/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
praveen:x:1000:1000:/home/praveen:/bin/bash
sshd:x:986:65534:sshd user:/run/sshd:/usr/sbin/nologin
rohit:x:1001:1001:,,,:/home/rohit:/bin/bash
tom:x:1003:1003:,,,:/home/tom:/bin/bash
bob:x:1004:1004:,,,:/home/bob:/bin/bash
```

Fields in /etc/passwd

1. Username

- User login name.
- Example: praveen

2. Password Placeholder

- Usually contains x.
- Actual password is stored in /etc/shadow.

3. UID (User ID)

- Unique identification number for the user.

- Example: 1000

4. **GID (Group ID)**

- Primary group identification number.
- Example: 1000

5. **GECOS**

- Stores user information such as full name.
- Example: Praveen Singh

6. **Home Directory**

- User's default working directory.
- Example: /home/praveen

7. **Shell**

- Default command interpreter.
- Example: /bin/bash

► **Structure of /etc/shadow File**

Command-

`sudo cat /etc/shadow`

Description

- Stores encrypted user passwords.
- Contains password aging and expiration information.

Importance

- Protects user passwords from unauthorized access.
- Improves system security.

```

praveen@praveenasus:~$ sudo cat /etc/shadow
[sudo: authenticate] Password:
root:!:20563:0:99999:7:::
daemon:!:20563:0:99999:7:::
bin:!:20563:0:99999:7:::
sys:!:20563:0:99999:7:::
sync:!:20563:0:99999:7:::
games:!:20563:0:99999:7:::
man:!:20563:0:99999:7:::
lp:!:20563:0:99999:7:::
mail:!:20563:0:99999:7:::
news:!:20563:0:99999:7:::
uucp:!:20563:0:99999:7:::
proxy:!:20563:0:99999:7:::
www-data:!:20563:0:99999:7:::
backup:!:20563:0:99999:7:::
list:!:20563:0:99999:7:::
irc:!:20563:0:99999:7:::
_apt:!:20563:0:99999:7:::
nobody:!:20563:0:99999:7:::
systemd-network:!:20563:0:99999:7:::1:
dhcpcd:!:20563:0:99999:7:::1:
messagebus:!:20563:0:99999:7:::1:
syslog:!:20563:0:99999:7:::1:
systemd-resolve:!:20563:0:99999:7:::1:
_chrony:!:20563:0:99999:7:::1:
landscape:!:20563:0:99999:7:::1:
polkitd:!:20563:0:99999:7:::1:
praveen:$y$j9T$vbuByBzu84pQdViSou0p20$Q1jhVy1.a0ixtbDqE735jGDcMKwMFheGVgTpqMDu
/V/:20609:0:99999:7:::
sshd:!:20611:0:99999:7:::
rohit:$y$j9T$BXTG7P5luzBi6E/Ag.7C60$loGmE1EtVKn7OVnIoz6cPLWsYwIyhFwPe1Wp1RquZ4
8:20615:0:99999:7:::
tom:$y$j9T$x1XqG6EeIPwPdG9KLGsAB.$SUSWLLMM9JRhXgBk01KuX5Zb.kc5i7F6MkRGMXVVL/:
20615:0:99999:7:::
bob:$y$j9T$9Fk51AQn3TRxpb6E1LUag1$e2gSiHjquCdXgblHh0U.ZvDjyjQh09PlpenWCJtfjVB:
20615:0:99999:7:::

```

► Structure of /etc/group File

Command-

cat /etc/group

Description

- Stores information about groups in Linux.
- Contains group names, GIDs, and group members.

```
praveen@praveenasus:~$ cat /etc/group
```

```
root:x:0:praveen
```

```
daemon:x:1:
```

```
bin:x:2:
```

```
sys:x:3:
```

```
adm:x:4:syslog,praveen
```

```
tty:x:5:
```

```
disk:x:6:
```

```
lp:x:7:
```

```
mail:x:8:
```

```
news:x:9:
```

```
uucp:x:10:
```

```
man:x:12:
```

```
proxy:x:13:
```

```
kmem:x:15:
```

```
dialout:x:20:
```

```
fax:x:21:
```

```
voice:x:22:
```

```
cdrom:x:24:praveen
```

```
floppy:x:25:
```

```
tape:x:26:
```

```
sudo:x:27:praveen
```

```
audio:x:29:
```

```
dip:x:30:praveen
```

```
www-data:x:33:
```

```
backup:x:34:
```

```
operator:x:37:
```

```
list:x:38:
```

```
irc:x:39:
```

```
src:x:40:
```

```
shadow:x:42:
```

```
utmp:x:43:
```

```
video:x:44:
```

```
sasl:x:45:
```

```
plugdev:x:46:praveen
```

Importance

- Helps manage permissions for multiple users.
- Simplifies access control and administration.

Task – 2: Managing Local Users

SOLUTION-

To create, modify, lock, unlock, and delete a local user account in Linux.

► 1. Create a User Called dev_user

Command

```
sudo useradd -m -s /bin/bash dev_user
```

```
praveen@praveenasus:~$ sudo useradd -m -s /bin/bash dev_user  
[sudo: authenticate] Password:
```

Explanation

- -m creates the home directory.
- -s /bin/bash sets Bash as the default shell.

► 2. Set a Password

Command

```
sudo passwd dev_user
```

```
praveen@praveenasus:~$ sudo passwd dev_user  
New password:  
Retype new password:  
passwd: password updated successfully
```

► 3. Modify User's GECOS Information

Command

```
sudo usermod -c "DevOps Engg." dev_user
```

```
praveen@praveenasus:~$ sudo usermod -c "Devops Engg." dev_user
```

Explanation

The **-c** option updates the comment (GECOS) field.

►4. Lock the User Account

Command

```
sudo usermod -L dev_user
```

```
praveen@praveenasus:~$ sudo usermod -L dev_user
```

Explanation

The account is locked and the user cannot log in.

►5. Unlock the User Account

Command

```
sudo usermod -U dev_user
```

```
praveen@praveenasus:~$ sudo usermod -U dev_user
```

Explanation

The account is unlocked and login is allowed again.

►6. Delete the User and Home Directory

Command

```
sudo userdel -r dev_user
```

```
praveen@praveenasus:~$ sudo userdel -r dev_user  
userdel: dev_user mail spool (/var/mail/dev_user) not found
```

Explanation

The **-r** option removes both the user account and its home directory.

Task – 3: Managing Local Groups

SOLUTION-

To create groups, add users to groups, verify memberships, and remove users from groups.

1. Create a Group Called developers

Command

```
sudo groupadd developers
```

```
praveen@praveenasus:~$ sudo groupadd developers
```

2. Add dev_user to the developers Group

Command

```
sudo usermod -aG developers dev_user
```

```
praveen@praveenasus:~$ sudo useradd dev_user  
praveen@praveenasus:~$ sudo usermod -aG developers dev_user
```

Explanation

- -a = append user to existing groups.
- -G = specify supplementary groups.

3. Verify Group Membership

Using groups Command

```
groups dev_user
```

```
praveen@praveenasus:~$ groups dev_user  
dev_user : dev_user developers
```

Explanation

These commands display the groups to which the user belongs.

4. Create a Secondary Group Called testers

Command

sudo groupadd testers

```
praveen@praveenasus:~$ sudo groupadd testers
```

5. Remove User from testers Group

Command

sudo gpasswd -d dev_user testers

```
praveen@praveenasus:~$ sudo gpasswd -d dev_user testers
Removing user dev_user from group testers
gpasswd: user 'dev_user' is not a member of 'testers'
```

Task – 4: Gaining Superuser Access

SOLUTION-

Difference Between su, sudo, and sudo -i

su

- su stands for Switch User.
- It switches from the current user to another user (usually root).
- Requires the target user's password.

sudo

- sudo allows a permitted user to run a single command as root.
- Requires the user's own password.
- More secure because it logs all privileged actions.

sudo -i

- sudo -i opens a root login shell.
- Loads root's environment and home directory.

- Similar to logging in directly as root.

►SUBTASKS-

1. Switch to Root Using su -

Command

sudo su -

```
praveen@praveenasus:~$ sudo su -
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.114.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Jun 14 15:50:16 UTC 2026

System load:  0.0              Processes:            35
Usage of /:   0.2% of 1006.85GB Users logged in:      0
Memory usage: 12%             IPv4 address for eth0: 172.20.
206.173
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster
   deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This message is shown once a day. To disable it please create the
/root/.hushlogin file.
root@praveenasus:~# whoami
root
root@praveenasus:~#
```

2. Add dev_user to the Sudoers File

Command

sudo usermod -aG sudo dev_user

```
praveen@praveenasus:~$ sudo usermod -aG sudo dev_user
praveen@praveenasus:~$ groups dev_user
dev_user : dev_user sudo developers
praveen@praveenasus:~$
```

3. Run a Privileged Command as dev_user

Switch to User

su - dev_user

```
praveen@praveenasus:~$ su - dev_user
Password:
```

Run Command

sudo cat /etc/shadow

```
$ sudo cat /etc/shadow
[sudo: authenticate] Password:
root:!:20563:0:99999:7:::
daemon:!:20563:0:99999:7:::
bin:!:20563:0:99999:7:::
sys:!:20563:0:99999:7:::
sync:!:20563:0:99999:7:::
games:!:20563:0:99999:7:::
man:!:20563:0:99999:7:::
lp:!:20563:0:99999:7:::
mail:!:20563:0:99999:7:::
news:!:20563:0:99999:7:::
uucp:!:20563:0:99999:7:::
proxy:!:20563:0:99999:7:::
www-data:!:20563:0:99999:7:::
backup:!:20563:0:99999:7:::
list:!:20563:0:99999:7:::
irc:!:20563:0:99999:7:::
_apt:!:20563:0:99999:7:::
nobody:!:20563:0:99999:7:::
systemd-network:!:20563:0:99999:7:::1:
dhcpcd:!:20563:0:99999:7:::1:
messagebus:!:20563:0:99999:7:::1:
syslog:!:20563:0:99999:7:::1:
systemd-resolve:!:20563:0:99999:7:::1:
_chrony:!:20563:0:99999:7:::1:
```

4. View the Sudo Log

Ubuntu

sudo tail /var/log/auth.log

```
praveen@praveenasus:~$ sudo tail /var/log/auth.log
2026-06-14T15:54:45.941683+00:00 praveenasus sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1005)
2026-06-14T15:54:45.945791+00:00 praveenasus sudo: dev_user : TTY=/dev/pts/0 ; PWD=/home/praveen ; USER=root ; COMMAND=/usr/bin/cat /etc/shadow
2026-06-14T15:54:45.950075+00:00 praveenasus sudo: pam_unix(sudo:session): session closed for user root
2026-06-14T15:56:22.242258+00:00 praveenasus su[2567]: pam_unix(su-l:session): session closed for user dev_user
2026-06-14T15:56:22.283374+00:00 praveenasus systemd-logind[170]: Session c7 logged out. Waiting for processes to exit.
2026-06-14T15:56:22.283706+00:00 praveenasus systemd-logind[170]: Removed session c7.
2026-06-14T15:56:32.353855+00:00 praveenasus systemd-logind[170]: Removed session c8.
2026-06-14T15:56:32.354624+00:00 praveenasus (sd-pam): pam_unix(systemd-user:session): session closed for user dev_user
2026-06-14T15:56:47.296821+00:00 praveenasus sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
2026-06-14T15:56:47.297380+00:00 praveenasus sudo: praveen : TTY=/dev/pts/0 ; PWD=/home/praveen ; USER=root ; COMMAND=/usr/bin/tail /var/log/auth.log
praveen@praveenasus:~$
```

